Do Fandoms Have a Voice? Tracing Stylistic Conventions across Fan Communities and Genre Norms

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Abstract

In this paper we employ stylistic and semantic analyses on a corpus of fanfiction to understand the differences in authorial voice across fan communities and genre norms. We compare the writing style of a random sample of fanfiction against five specific fan communities. This allows us to compare the specifics of individual fans' and fan communities' writing styles to map how the writing style of individual fans collectively create the writing style of the community. We measure both the differences between communities as well as the variation within communities over time to assess the degree of stability of communal authorial voices. Additionally, we compare the writing style of four different Maturity Ratings to reproduce findings from previous research that finds genre norms to influence the linguistic style of fanfiction. Based on a factor analysis of 71 measures, we score texts along 10 explainable factors that are then used to compare and contrast groups via coefficients derived from a multinomial logistic regression classifier. Our results indicate that fandoms have distinct writing styles that arise after a period of conventionalization and subsequently ebb and flow across a baseline writing style range. Similar for Maturity Ratings, we find General Audience texts and Explicit texts to be clearly differentiated; however, model performance indicates that Maturity Ratings are not as easily separable as fandoms when it comes to our measures of stylistic variation.

Keywords: fanfiction, stylistics, authorial voice, statistical modeling, natural language processing

1 Introduction

Fanfiction is fiction texts written by and for fans that center around characters or worlds from a pre-existing story, which are published online for free [21; 39]. They are often written serially, meaning one chapter is posted at a time. Most platforms allow for reader-writer interactions via comments and reviews on individual chapters. These ongoing reader-writer interactions make fanfiction texts communal, co-creative endeavors, where the norms of the group shape the texts that are written, which in turn shape the norms and values of the communities [6]. Although fanfiction is acknowledged as deeply communal with a clear temporal element, there is little existing literature exploring temporal trends of stylistic negotiation between authors and the communities they write within. Previous work has identified tendencies towards the convergence of authorial voice among group peers, though this work remains broadly focused on domain-level divisions, such as academic works, social media posts, and literature [8; 11; 16].

Instead, we take a more directed approach, exploring textual works from tightly-knit fan communities, or *fandoms*, in order to observe how more personal peer associations impact group writing culture. In doing so, the identification of fandom specific and agnostic patterns of stylistic reorganization may provide key insights as to how authors together reshape their community that, in turn, shapes them. Increased similarity in authorial voice has been found to relate to the likelihood of fellowship amongst authors, and this fellowship has conversely been associated with a convergence in authorial voice [27; 30]. We expect to substantiate these findings by qualifying the interplay of author and fandom in defining one another.

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2 Related Work

2.1 Stylometry, Authorial Voice, and Community Norms

Originally defined as an *inherent signature* of one's work, authorial style has since been renegotiated as a textual artifact of emergent processes that are context-driven, cognitively impacted, and dynamic [4; 14; 18; 57]. This understanding serves as the basis for our definition of authorial voice; it is the amalgamated profile of many textual expressions, informed by the author(s) goals, experience, contextual restraints, and internal state. This re-conceptualization has been accompanied by the popularization of stylometric methods as a more objective approach to understanding authorial style through quantitative measures [25; 32; 44]. It is important to note that, from one work to the next, subsequent texts share many linguistic similarities [1; 23; 47]. This framing suggests, however, that these similarities are not the externalization of some internal trait, but instead the result of congruent context and external pressures [16; 29; 46; 53].

Evidence already supports the use of stylometric measures to explore community linguistic dynamics, suggesting a shift and stabilization towards community norms over time [10; 41]. Furthermore, batteries containing a large quantity and diversity of stylometric measures have succeeded in grouping online communities based on their linguistic fingerprint [26; 52]. This work, however, characterizes the author simply as a recipient of community conventions rather than an active consumer, influencer, and producer of creative norms. Thus, the implementation of stylometric tools here will serve as a means for directly quantifying the interaction between individual authorial voice and these norms.

2.2 Computational Fanfiction Research

Although fan scholarship has its roots in qualitative, ethnographic research [3], there has been a growing interest in computational fanfiction research [21]. A central theme of this research is the stylistics of fanfiction as it pertains to fan engagement or, in other words, what is the best or most popular style of fanfiction [21]. These studies find popular or appreciated fanfiction to be longer, with a wider vocabulary, and with more indicators of direct speech such as second person pronouns [31; 42; 49].

However, more recent research has shown that correlating the style of a fanfiction text with fan engagement metrics overlooks central aspects of the context in which fanfiction is created and shared. For one, specific fan communities have been found to have distinct textual styles which mirror the style of the textual source material [20]. Additionally, there are genre conventions that influence the writing style of fanfiction, such as Maturity Ratings – the metadata tags that are added by the author to denote both the degree of explicit content in the text as well the intended audience. On fanfiction platforms, texts from differing Maturity Ratings have been found to have distinct textual styles [22]. The distinct writing styles of these ratings could imply genre conventions that supersede and are simultaneously imposed on the communities and their respective writing styles. Additionally, the features correlated with success change depending on which measures of success are used [19].

Studies focusing on diachronic change in fanfiction communities have focused primarily on how power dynamics between characters change over time using the Riveter tool developed by Maria Antoniak [2; 38; 58]. Riveter is a dictionary-based tool that works by identifying words relating to characters in a text, and then assigns power and agency scores to characters based on the words associated with them [2]. Riveter thus approximates how characters are positioned in relation to each other in the text through word choice. Yang and Pianzola [58] compared the portrayal of eight different fanfiction couples and consistently found the difference in power scores to become more stable over time, as well as converging towards zero. In other words, not only did in-community variance decrease, it also moved towards a more equal portrayal of the couples. In

contrast, Neugarten [38] assessed the power differences between Greek myth characters Hades and Persephone in short fanfiction texts. Neugarten [38] found the difference in power scores between Hades/Persephone to change in waves with periods where Hades was portrayed as more powerful, followed by periods where Persephone was positioned as more powerful.

The question of diachronic change in fanfiction communities is central due to the ongoing community interactions that shape the texts over time. We find it a timely contribution to further nuance these findings from previous research using updated, fine-grained methodological changes and new use-cases. As such, we investigate whether there are fandom-specific writing styles, and how such a style within a community might arise, become ingrained, and shift through the conventionalization of individual authors' voices.

3 Methods

To understand the influence of the community on fans' expressive profiles, we use a collection of fanfiction corpora from several fandoms as well as a control corpus of randomly sampled fanfiction. We measure the stylistic profile of each fanfiction, after which we use statistical analysis to gauge the specific style of each fan community and the variance within them.

3.1 Dataset

The dataset is a collection of fanfiction from the platform *Archive of Our Own*[1], also known as AO3, which currently hosts over 18 million works from over 78,000 fandoms. Our corpus consists of six different random samples of around 3,000 texts each, totaling in 18,488 unique works. Five of these samples originate from specific fandoms from AO3, while the sixth sample is a random sample of texts from AO3 to function as a control group.

We use the data from [19] and [40], which contains data from the fandoms *Harry Potter* (HP), *Lord of the Rings* (LOTR), and *Percy Jackson and the Olympians* (PJ). Common to these three fandoms is the fact they are each based on a specific fantasy novel series. Worth noting as well is that each of these novel series have been adapted to the screen as movies and/or TV shows. Our fourth group is known as *Real Person Fiction* (RPF), and is a sample of texts from the dataset in [59]. These texts, in contrast to the previous groups, are not based on a specific, textual source material. Instead, it is a collection of fandoms that all center around real-life people such as WWE wrestlers, football players, actors, pop stars, etc. The fifth fandom is based on the horror podcast series *The Magnus Archives* (MAG). The source material is a fictional audio horror drama that is a collection of short stories held together by an overarching narrative. This group was included to have a fiction-medium that was non-textual in contrast to the first three fandoms. Finally, as a control corpus, we have a random sample of fanfiction published on AO3 between January 2023 and January 2026. These texts center around 1710 different fandoms, and allow for both crossovers (where multiple fandoms are included in a single text) and original work (texts that are original piece of fiction posted on AO3 under the same affordances as fanfiction).

The texts were collected in accordance with the site's terms of service, and were treated as sensitive data in accordance with GDPR due to ethical considerations with regards to fans' privacy [7]. Texts that were either improperly tagged as English (630 works) by AO3 or within the bottom 5th percentile in text length (819 works) were excluded as they often contained romanized foreign-language text or metadata on other works, respectively. This resulted in a final corpus of 16,966 fanfiction, detailed more comprehensively in table 1. Additionally, for the Maturity Rating analysis, we removed texts that were tagged as 'Not Rated', as we are interested in the cases where authors have made a genre-specific choice for their text.

[1] https://archiveofourown.org/

Fandom	N.	Oldest text	Newest text	Mean N. words	SD words
HP	2,971	2002-01-19	2023-12-28	13,144	33,090.32
LOTR	2,973	2002-02-04	2023-11-17	11,853	31,147.47
PJ	2,977	2009-07-22	2023-11-20	8,585	18,722.31
RPF	2,609	2004-06-21	2024-03-10	6,844	21,298.32
MAG	2,972	2017-01-07	2024-09-03	6,688	17,369.01
Random	2,461	2022-12-31	2026-01-09	10,933	36,775.17

Rating	N.	Oldest text	Newest text	Mean N. words	SD words
General Audiences	5,300	2002-01-19	2026-01-05	4,418	11,279.2
Teen and Up Audiences	5,473	2000-04-30	2025-12-29	9,922	25,984.6
Mature	2,097	2002-09-18	2026-01-03	18,400	43,605.0
Explicit	2,551	2002-02-04	2025-12-29	14,308	36,445.8
Not Rated	1,545	2003-01-19	2025-12-24	7,584	19,441.7

Table 1: Description of the dataset group by both Fandom and Maturity Rating, after cleaning.

3.2 Stylometric Feature Extraction

Our analysis builds on the aforementioned success of large stylometric batteries in assessing the linguistic fingerprint of online communities [26; 52]. To construct a stylometric battery that holistically represents style, an inductive approach was used to assess the existing literature. For similar tasks such as authorship attribution, four feature families have been found to accurately predict unique authors and were thus included: lexical, syntactic, structural, and character-level features [12; 13; 43; 50]. As this study extends beyond purely classification towards a more holistic profiling of authors, two additional families were included: affective and semantic features [36; 51]. Despite their exclusion in common stylometric studies, they are frequently included in close-reading stylistic work. Once these families were identified, a literature review of each family was undergone to identify the most prevalent metrics used in existing literature. These stylometric domains, categories, and features used are outlined in detail in table 2.

Character-level measures were calculated by counting instances of characters of interest and dividing them by token count to retrieve a text-level character per token ratio [5]. The Python package TextDescriptives [15] was employed to compute three types of measures: structural measures of readability, repetition, and document size; syntactic measures of parts-of-speech and dependency distance; and lexical measures of word-form complexity and richness. These measures were further supplemented by syntactic complexity measures derived from *en_core_web_md* spaCy embeddings [35], as well as lexical measures of entropy using smoothed n-gram Shannon entropy [9; 48], and registered using an age-of acquisition [28] lexicon. Three further lexicons, the NRC VAD Lexicon v2 [34], NRC Emotion Intensity Lexicon v1 [33], and Brysbaert Concretness Ratings [37], were used to score affective dimensions, discrete emotions, and textual concreteness, respectively. Lastly, principle component analysis and k-means clustering of text-level E5-small sentence-transformer embeddings were used to calculate semantic measures of topic strength, volume, clustering, dispersion, coherence, and trajectory [24; 45; 54; 55]. Each individual fanfiction received a score for all 71 metrics which were then used for further factor analysis [2].

[2] Anonymized code is available for review: paper-specific analysis code at `https://anonymous.4open.science/r/CHR-Fanfiction-Paper/README.md`, and the feature-extraction pipeline at `https://anonymous.4open.science/r/Expressive-Profile-Pipeline/README.md`. Raw AO3 texts are not shared, as they are treated as sensitive personal data under GDPR.

Family	Category	Metrics
Character-Level	Punctuation Style	Rate per token of comma, colon, question, exclamation, ellipsis, parenthesis, and quotation
Structural	Document Scale	SD of sentence length (tokens)
	Readability	Coleman-Liau, SMOG, and RIX scores
	Repetition	Character fraction of duplicate lines, duplicate 5- and 8-grams, and top 2-, 3-, and 4-grams
Lexical	Lexical Richness	Alphabetic-token and out-of-vocabulary ratios, and proportion of unique tokens
	Word Form Complexity	Token length (SD) and syllables per token (mean, SD)
	Lexical Register	Age of acquisition (mean)
	LM Entropy	Windowed unigram entropy in nats (SD) and perplexity (mean)
Syntactic	POS Distribution	Proportion of tokens tagged as noun, verb, adjective, adverb, pronoun, determiner, adposition, auxiliary, particle, interjection, and coordinating and subordinating conjunctions
	Syntactic Complexity	Coordination and subordination ratios, proportion of complex sentences, dependents per head (mean), and parse-tree depth (mean, SD)
	Dependency	Dependency distance (SD) and proportion of adjacent dependency relations (mean, SD)
Affective	Affect Dimensions	Valence, arousal, and dominance (mean, SD)
	Concreteness & Imagery	Word concreteness (mean, SD)
	Discrete Emotions	NRC scores for anger, fear, disgust, sadness, joy, trust, anticipation, and surprise, plus overall emotion diversity
Semantic	Topic Strength & Volume	First-principal-component topic strength and semantic volume (log-determinant)
	Clustering	Maximum embedding-cluster share
	Dispersion	Semantic dispersion around the centroid
	Coherence & Trajectory	Segment transition variability, start-to-end semantic drift, and maximum pairwise semantic range

Table 2: Stylometric families, categories, and features applied on the fanfiction corpus, totaling in 71 metrics across 6 families and 18 categories, ordered from character-level to semantic.

3.3 Exploratory Factor Analysis

As this study's scope is more broadly focused on group style dynamics as opposed to characteristics of individual authors, an exploratory factor analysis (EFA) was performed on the data [56]. With EFA, the aim is to identify groups of measures that correlate and simplify them into single factors that represent latent relationships in the data. Doing so allows us to discern overarching factors that impact authorial voice without the bias of a single a priori hypothesis present in other forms of factor analysis [17]. As a result, each fanfiction receives a score for all identified factors, alongside individual stylometric measures, to better explain how certain factors present within a given fanfiction.

Before performing the EFA, normality was improved by Yeo-Johnson, transforming positively skewed metrics and capping outliers to ± 3.5 z-scores from the mean. After addressing these assumptions, the data was determined to be suitable for EFA using both the Bartlett's test of sphericity ($\chi^2(2{,}701) = 1{,}138{,}322.95$, $p < .001$) and the Kaiser–Meyer–Olkin measure of sampling adequacy ($p = 0.849$), signifying that these metrics were sufficiently interrelated for grouping using EFA. Lastly, before fitting, we decided upon grouping metrics into 10 latent factors based on a combination of parallel analysis and the amount of cumulative variance explained with each additional factor. The EFA was conducted using minimum residual extraction to account for remaining non-normality in the transformed data and oblim rotation, as these features are conceptually related. All 71 stylometrics were grouped into one or more of the ten factors, whose loadings above 0.30 were then assessed to determine what latent constructs each feature represents. Table 3 presents the name for each factor, the metrics that make it up, the variance explained by a single factor, and

the total amount of variance explained with each additional factor.

Factor	Name	% Var.	Cum. %	Top Loadings
F1	Syntactic Complexity	8.5	8.5	Subordination ratio (+.97); proportion of complex sentences (+.85); parse-tree depth, mean (+.82); SD of sentence length (+.79); coordination ratio (+.74); +7 more
F2	Lexical Sophistication	6.6	15.1	Syllables per token, mean (+.90); syllables per token, SD (+.89); token length, SD (+.78); SMOG (+.62); age of acquisition, mean (+.61); +5 more
F3	Negative Affect	6.6	21.7	Fear (+.85); anger (+.83); sadness (+.79); disgust (+.77); valence, SD (+.75); +5 more
F4	Lexical Richness	5.8	27.5	Topic strength (+.93); proportion of unique tokens (+.92); semantic volume, log-determinant (+.86); maximum embedding-cluster share (+.84); duplicate 5-gram fraction (−.61); +1 more
F5	Repetitiveness	5.5	33.0	Top 3-gram fraction (+.71); alphabetic-token ratio (−.61); top 2-gram fraction (+.58); top 4-gram fraction (+.58); out-of-vocabulary ratio (+.55); +9 more
F6	Concreteness	5.2	38.2	Word concreteness, mean (+.81); auxiliary, prop. (−.78); word concreteness, SD (+.75); arousal, SD (+.52); noun, prop. (+.46); +5 more
F7	Conversationality	5.1	43.3	Dependents per head, mean (+.68); interjection, prop. (+.59); question rate (+.54); determiner, prop. (−.50); noun, prop. (−.48); +6 more
F8	Positive Affect	3.7	47.0	Joy (+.85); trust (+.66); anticipation (+.57); valence, mean (+.56); valence, SD (+.48); +2 more
F9	Narrative Drift	3.5	50.6	Maximum pairwise semantic range (+.79); start-to-end semantic drift (+.67); semantic dispersion (+.66); segment transition variability (+.59)
F10	Structural Variability	3.2	53.8	Dependency distance, SD (+.62); verb, prop. (−.58); SD of sentence length (+.44); parse-tree depth, SD (+.35); pronoun, prop. (−.33); +3 more

Table 3: Ten-factor exploratory factor analysis, which accounts for 53.8 % of cumulative variance. For each factor, its label, amount of variance it explains, cumulative variance with each additional metric, highest-loading metrics, and their relationship to the factor are reported.

When a factor score for a text increases, this signifies that the given fanfiction relates more closely to the assigned factor label. For example, texts with a higher *Concreteness* (F6) score are more concretely written and thus, they likely display an increased mean word concreteness, decreased proportion of auxiliary verbs, and so on. All but two of the ten factors contain metrics from two or more stylometric families, which suggest that our analysis is capturing latent structures of authorial voice and not simply differences in how measures are collected, processed, or scored. Moreover, the two factors composed of a single family, *Positive Affect* (F8) and *Negative Affect* (F3), further support the interpretablity of the EFA as they both neatly separate from one another and only include other linguistic measures that theoretically favor a certain emotional pole. The full pattern matrix, included in appendix A, contains the mapping of factor loadings for each included metric. With these identified factors of authorial voice in fanfiction, we have clear interpretablity domains defined by the works themselves through which community creative norms can be understood.

3.4 Analytical Approach

In the following section, we outline the analysis we performed in order to assess whether different fandoms and Maturity Ratings exhibit different stylistic profiles, as well as how these fandoms' profile might change over time. We thus address the following three research questions with our analysis:

1. Do different fandoms have their own unique authorial voice?

2. Does this group profile converge, diverge, or remain the same over time?
3. Do different Maturity Ratings have their own genre profiles?

3.4.1 Distinctiveness across Fandoms

We performed a multinomial logistic regression with the aim of assessing the efficacy of utilizing the 10 factor scores produced by the EFA in predicting the fandom of a given fanfiction. To assess the predictive performance, a 5-fold cross-validation was implemented in which four folds, or 80% of the fanfictions in the corpus, were used to train the model, which was then used to predict the fandom membership of the fifth fold, or the withstanding 20% of fanfictions. This was done four more times so that each fold could be tested. Comparisons were then made between a works true fandom membership versus its predicted membership and its performance was assessed in the related results section.

In addition to identifying possible differences in style, the regression provides valuable information via model coefficients as to which factors most directly influence which fandom a work was predicted to be a part of. Using the direction and magnitude of a coefficient, we can determine which stylistic features most directly impact fandom-mediated differences in authorial voice. This, in conjunction with an assessment of the performance of the model, allows us to assess whether notable fandom mediated differences in authorial voice exist, as well as what features of the text may be driving these differences.

3.4.2 Fandom Stability and Convergence over Time

In order to investigate the change over time across fandoms, we calculated the rolling average and rolling standard deviation for each factor over time. In other words, we divided the dataset into fandoms, sorted by date, and then calculated the rolling average and rolling standard deviation with a window size of 30 for each factor.

To then assess how the factor values change over time across groups, we plotted the two arcs. In order to better compare groups, we min-max normalized the publication date. With Random as a baseline, we compared the rolling average arc and rolling standard deviation arc across fandoms. The rolling average will be used to indicate whether and how the fanfiction texts' expressive profiles change over time, and the rolling standard deviation will indicate whether the groups have increasing or decreasing variance over time.

3.4.3 Distinctiveness across Maturity Ratings

As an exploratory analysis, as well as to try and replicate previous findings by Jacobsen [22], we test whether different Maturity Ratings have unique stylistic profiles. Once more, we performed a multinomial logistic regression with 5 fold cross-validation that, in this case, predicted the Maturity Rating using the 10 EFA factors and saved model coefficients, predictions, and performance metrics. This allows us to both assess the predictive power of the factor scores for Maturity Ratings, as well as evaluate how authorial voice differs across Maturity Rating categories.

4 Results

4.1 Distinctiveness across Fandoms

Of note, first and foremost, is the overall performance of the multinomial logistical regression in predicting the fandom of a given fanfiction. Across 5 folds, the model correctly assigns the fandom around 40% of the time (See table 4). This becomes more informative when considering the model's Macro ROC-AUC of 0.737 ± 0.003, which tells us that the model correctly separates in-fandom examples from out-of-class examples in 74% of comparisons.

Class	Precision	Recall	F1
HP	0.349 ± 0.020	0.269 ± 0.010	0.303 ± 0.012
LOTR	0.439 ± 0.011	0.488 ± 0.026	0.462 ± 0.016
MAG	0.406 ± 0.022	0.480 ± 0.019	0.440 ± 0.017
PJ	0.366 ± 0.028	0.332 ± 0.025	0.348 ± 0.021
RPF	0.339 ± 0.026	0.311 ± 0.016	0.324 ± 0.013
Random	0.457 ± 0.012	0.511 ± 0.022	0.482 ± 0.010
Macro avg.	0.393 ± 0.003	0.399 ± 0.002	0.393 ± 0.003

Table 4: Per-Class Performance across 5 folds (mean $\pm$ std). Accuracy $= 0.397 \pm 0.002$; Macro ROC AUC (one-vs-rest) $= 0.737 \pm 0.003$.

Together, this suggests that despite the fact that the model does not perform astonishingly well at assigning fandom – though better than chance – the model often considers the true fandom likely to be the correct prediction even if it ultimately chooses another fandom. This supports the notion that there is distinct information held within the derived factors that informs the models ability to distinguish fandom.

When looking at per-class model performance presented in table 4, it becomes clear that the model is best at identifying fanfiction texts from the LOTR (F1 = 0.46) and MAG (F1 = 0.44) fandoms. This is in contrast to HP (F1 = 0.31), PJ (F1 = 0.35), and RPF (F1 = 0.32), which present more of a struggle for the model. Interestingly, these three fandoms shared recall scores that were lower than their precisions scores, whereas LOTR and MAG displayed the opposite. This suggests that LOTR and MAG may have a unique profile and are thus over-predicted, absorbing false positives from other classes, while the less distinct fandoms become confused with one another (for the confusion matrix, see appendix B).

The factor-profile plots included in figure 1 further inform this interpretation. Upon visual inspection, the HP and PJ plots are generally more rounded, with fewer factor means that deviate widely from the population mean. Conversely, the shapes produced in the LOTR and MAG plots are noticeably more jagged and farther deviated from the population mean. These four plots support the straightforward interpretation that a more widely deviated and directionally varied profile is easier to categorize. Looking at the RPF and Random plots, however, this interpretation is placed into question. Both plots are uniquely shaped with dimensions that differ considerably from the population, but RPF is one of the least identifiable fandoms, whereas Random is the most correctly identified (pseudo-)fandom in the model (F1 = 0.48). Thus, looking more closely at the EFA factor coefficients and their impact on predictions becomes necessary.

Figure 2 below supplements the above plots by presenting the degree to which a given factor helps to inform the prediction of a given fandom, with larger factor coefficients symbolizing a stronger relationship, either positive or negative, between a factor and the likelihood of a given fandom. The coefficients can also be seen in table 4 in appendix B.

For RPF, it is signified by a fairly even share of information across multiple different factors. The authorial voice of RPF is differentiated mainly by decreased usage of *Repetitiveness* (F5, β = -0.42) along with increased *Lexical Richness* (F4, β = 0.46) via broadness of topics and other semantic information and *Lexical Sophistication* (F2, β = -0.32) via sentence length and complexity. All three factors share a similar magnitude of relationship to the fandom label, with the other six informative dimensions remaining fairly evenly distributed in magnitude as well.

Random, conversely, is defined mainly by significantly decreased *Repetitiveness* (F5, β = -1.17), which is much farther removed from the next two most important features of Random fanfiction texts, decreased *Syntactic Complexity* (F1, β = -0.47) and increased reliance on signifiers



Figure 1: Mean factor-score profile by fandom (10 EFA factors; grey band = ± 3 SE). Values are offset so the dashed red circle marks the true zero — outside it is above the population mean; inside is below.

of *Conversationality* (F7, β = 0.40), as well as the seven other informative factors. Taken together, this suggests fandoms that are easy to predict are not simply those with the most factors that differ significantly from the population mean, but possibly instead those that have one or two clearly defined features that stand out from the rest, acting as clear, non-complex markers of fandom membership.

Looking at the other four fandoms, this interpretation becomes more clear. The two next best predicted fandoms, LOTR and MAG, share a similar distribution of coefficient values as Random. LOTR, for example, is primarily defined by a decreased use of *Conversationality* (F7, β = -0.76),

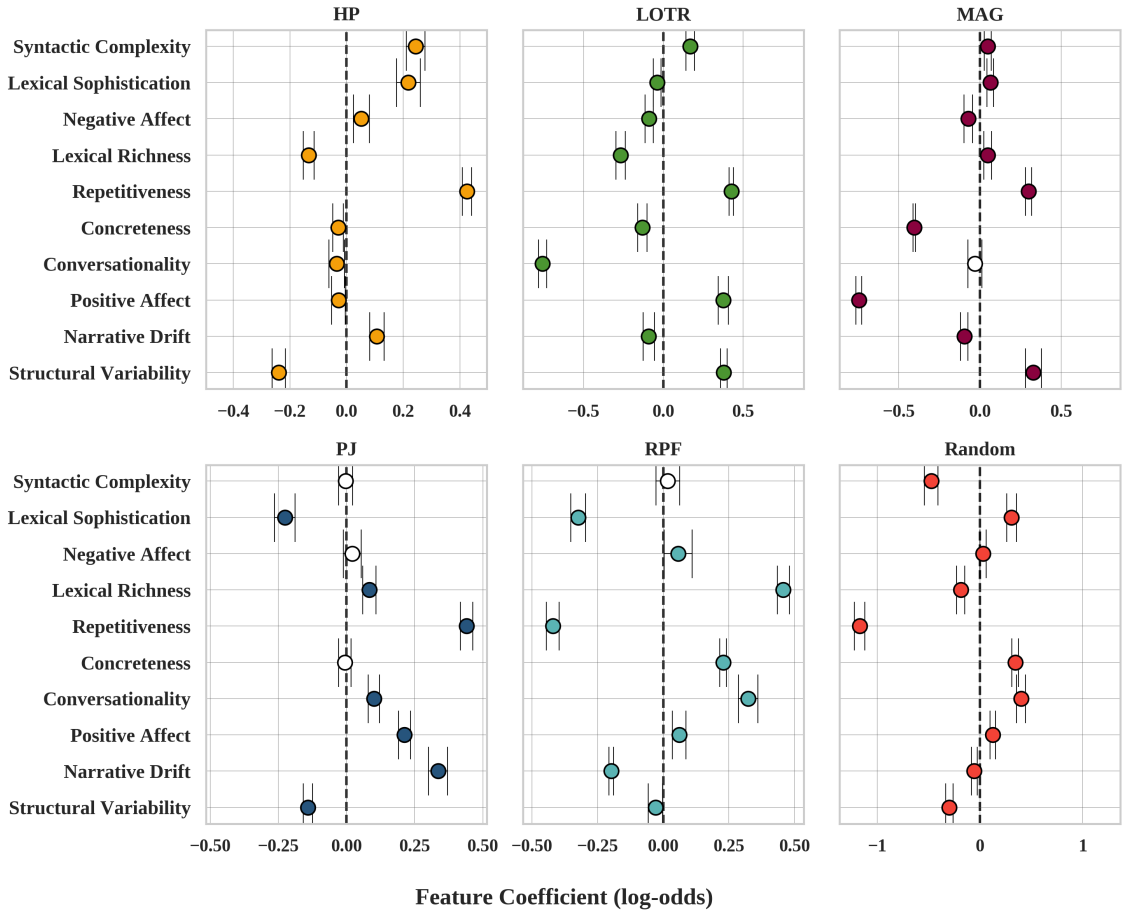


Figure 2: Coefficient plot from multinomial logistic regression predicting fandom from 10 EFA-derived stylistic factors. Dots show mean coefficients (log-odds) across 5 cross-validation folds. Filled dots indicate the mean is 3 or more standard deviations from the mean, while hollow dots indicate the effect is not stable across folds. Error bars are ±3 SD across folds.

with increased *Repetitiveness* (F5, β = 0.43), *Positive Affect* (F8, β = 0.38), and *Structural Variability* (F10, β = 0.38) constituting the next highest defining features of the remaining nine. As with Random, LOTR presents a clear identifying factor that signifies LOTR membership; this is also corroborated by previous research identifying LOTR fanfiction as minimally conversational [20] and more positive [39]. MAG, to a lesser extent, shows this same trend, with decreased *Positive Affect* (F8, β = -0.75) proving most relevant, with *Concreteness* (F6, β = -0.40), *Structural Variability* (F10, β = 0.33), and *Repetitiveness* (F5, β = 0.30) the next closest defining features of MAG fanfiction texts.

HP, on the other hand, does not have a single factor that is over a magnitude of $|0.5|$. Instead, increased *Repetitiveness* (F5, β = 0.43), *Syntactic Complexity* (F1, β = 0.24), and *Lexical Sophistication* (β = 0.22) best signify HP fanfiction texts. The same is true of PJ, with *Repetitiveness* (F5, β = 0.44), *Narrative Drift* of semantic meaning across the text (F9, β = 0.34), and *Lexical Sophistication* (F2, β = -0.23) defining fanfiction texts with this label. Consistent with above, neither fandom carries a clear factor that uniquely defines the fandom from others. Moreover, the magnitude of the remaining factor coefficients is also fairly small in relation to the reported coefficients of the other fandoms. As such, it becomes clear that a fandom profile with a significant, uniquely defining factor in a well performing model is most easily identifiable.

It is evident from the visual profiles of the 10 EFA factors that fandom membership is indicative of unique authorial voice, as the combinations of features that meaningfully signify a given group are varied in both order and magnitude. However, with the inclusion of both model performance and coefficients, this becomes more nuanced. In addition to the order and magnitude of factors, the distribution of these factors is important when trying to classify a fandom from the others. Specifically in classification, it appears that a profile with a clear, interpretable ”fingerprint” is much easier to classify than a profile that, though equal in magnitude, is more integrated and nuanced.

In this way, the best predicted fandom should not be interpreted as a proxy for the one with the most prominent display of authorial voice, but instead the profile with the clearest interpretablity. Detectability and distinctiveness are not the same thing. A fandom is easiest to classify not when its authorial voice is strongest, but when its stylistic profile is most clearly differentiated.

This may be explained by the relation of the fanfiction and fandoms to their source text, as fanfiction texts from these same fandoms have been found to mirror closely the style of the original work [20]. LOTR and MAG may both be distinctive enough in authorial voice from the remaining fandoms to produce clearly differentiated profiles, whereas HP and PJ are both closer in genre, audience, and narrative conventions, and thus may be more overlapping in style. Random and RPF may perform better as they are not inhibited by features inherent to the source text – they are both an amalgamation of different communities. Whether these differences in authorial voice persist or shift systematically as a community matures, however, remains an open question that is assessed in the following section.

4.2 Fandom Stability and Convergence over Time

In figure 3 we visualize the rolling average for each factor across fandom and publication date. As explored in RQ1, the fandoms differ on the baseline level of each factor, most clearly visible for factors 2, 5, 7, and 8.

Another pattern we observe in the rolling average visualization is a period of convergence happening early on within the fandoms. Once this period of convergence is complete, the average ebbs and flows from the baseline. In other words, for the five specific fan groups where data is collected across their lifetime on AO3, each group seems to start with a clearer, consistent trend either upwards or downwards. Once the group stabilizes towards a baseline though, it goes through shorter periods of stylistic trends, almost as if the group members act and react to each other’s style choices. Once a writing style becomes prevalent, there is a pull towards the opposite style, but still centered around the same baseline level of that style.

As a sanity check, we find it positive that we observe only the ebb and flow in the random sample and not the period of stabilization. Since these texts are not part of a specific group, we would not expect group convergence, and simultaneously, the random sample only covers a short period of time, and does not in the same way cover the beginning of an arc. These findings and interpretations are in line with previous research, where the power differences between Hades and Persephone as portrayed in fanfiction also displayed clear ebbs and flows [38].

Turning to the rolling standard deviation visualized in figure 4, we see a surprising pattern of findings. Instead of seeing a decreasing level of variance over time as groups converge – which is what we expected given the findings in [58] – we instead see first a rising trend for several factors and groups, which indicate a period of increased variance. Afterwards, the standard deviation again ebbs and flows like the rolling average.

It is not clear whether the bumps in the standard deviation come before or after the trends in the rolling average. In other words, we cannot tell whether periods of greater variance are a product of the style change in a specific direction, or whether the periods of greater variance happen before a change in writing style. More research and more sensitive modeling would be necessary to assess

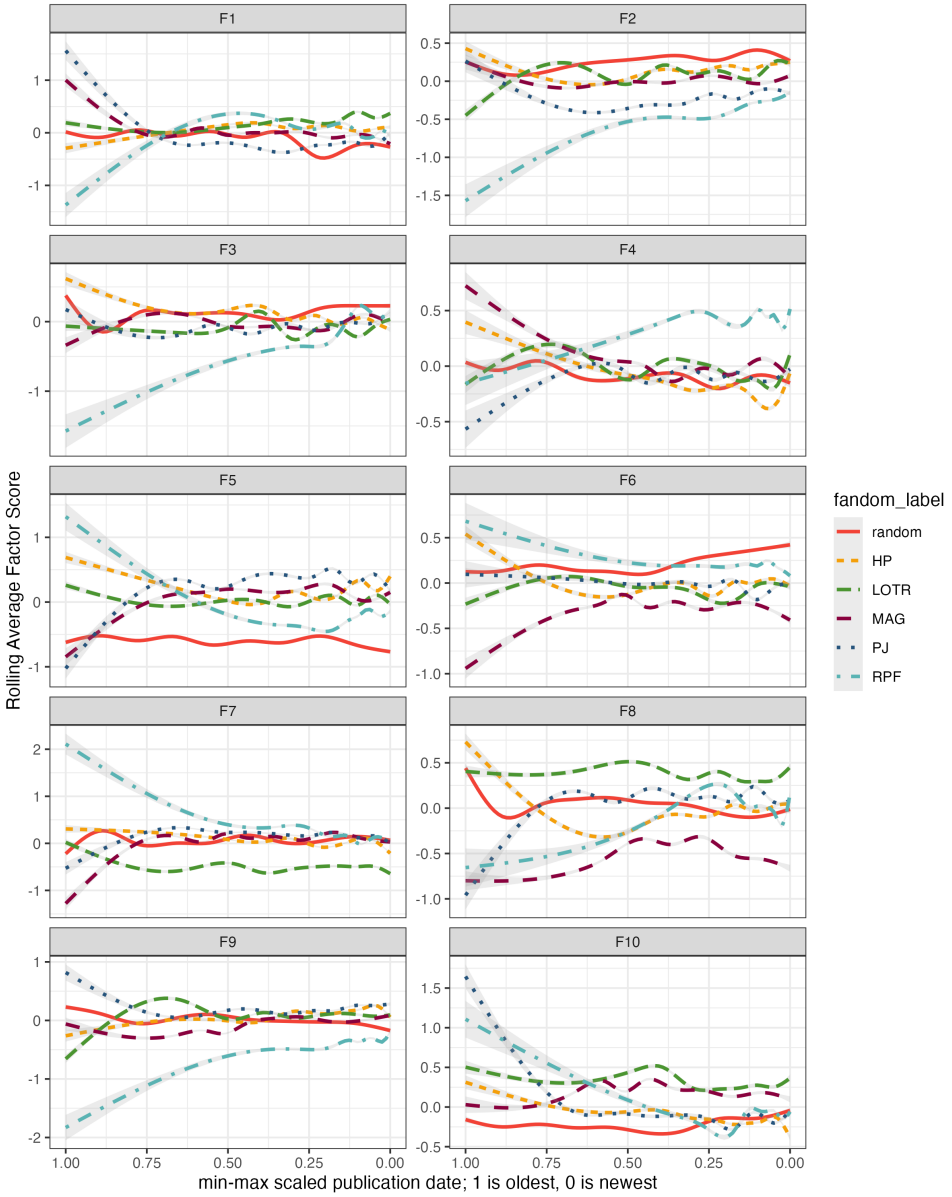


Figure 3: Rolling average (window size=30) of factor scores across fandoms and publication date. Publication date is min-max normalized, where 1 is oldest and 0 is newest. Shaded areas indicate 95% confidence intervals, and are particularly broad in time periods with scarce data (i.e., the window must cover a greater time period than more populated time periods).

the relation between these two patterns. What we can determine, however, is that there seems to be a consistent negotiation and re-negotiation of style between authors in a fandom that, though it shifts, remains within a fairly stable ”border” of style unique to that community.

4.3 Distinctiveness across Maturity Ratings

In table 5, we report the classification performance metrics for the Maturity Rating prediction task. Overall, the model performance is not great. With a mean accuracy across folds at just below 50% and an ROC-AUC of 0.45, the model does perform better than chance but it still seems to have a difficult time predicting the correct labels. Across the 5 folds, we find General Audience

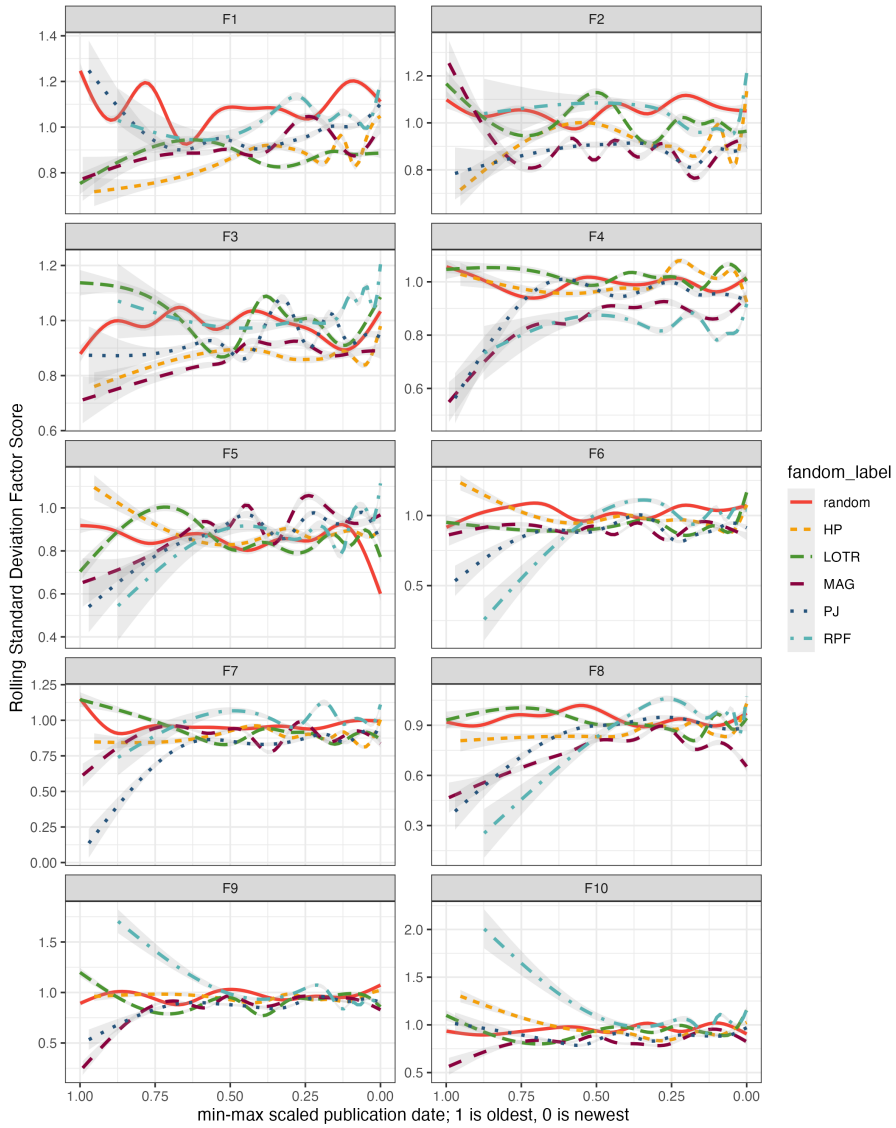


Figure 4: Rolling standard deviation (window size=30) of factor scores across fandoms and publication date. Publication date is min-max normalized, where 1 is oldest and 0 is newest. Shaded areas indicate 95% confidence intervals.

fanfiction to have the best performance with a mean F1-score of 0.58, while Mature has the worst performance with a mean F1-score of 0.014. The main driver for this poor performance is the recall for Mature, which is only 0.0073, while the precision for Mature is 0.61. In other words, when the model predicts a text as Mature, it is usually right, but it is very conservative in its use of the label.

Looking at the factor score-profile for the different Maturity Ratings (figure 5), it is clear that the most notable distinction between Maturity Ratings is the difference between General Audiences and Explicit. The factor profile for Explicit has high scores for *Concreteness* (F6) and *Syntactic Complexity* (F1) while having low scores for *Repetitiveness* (F5), *Lexical Richness* (F4), and *Structural Variability* (F10). Almost completely opposite, General Audience texts have a high score for *Lexical Richness* (F4), *Positive Affect* (F8), and *Structural Variability* (F10), while having a low score for *Negative Affect* (F3).

As such, the profiles of fanfiction for General Audiences versus Explicit are fairly inverse, as the relationship between a given factor and Maturity Rating shifts sign completely for six of the

Class	Precision	Recall	F1
General Audiences	0.527 ± 0.015	0.571 ± 0.012	0.548 ± 0.011
Teen and Up Audiences	0.433 ± 0.016	0.59 ± 0.0087	0.499 ± 0.012
Mature	0.614 ± 0.22	0.0073 ± 0.0057	0.0143 ± 0.011
Explicit	0.482 ± 0.0089	0.414 ± 0.020	0.445 ± 0.014
Macro avg.	0.514 ± 0.05	0.396 ± 0.009	0.377 ± 0.008

Table 5: Per-Class Performance across 5 folds (mean $\pm$ std). Accuracy $= 0.475 \pm 0.008$; Macro ROC AUC (one-vs-rest) $= 0.450 \pm 0.008$.

ten factors from General Audiences to Explicit (*Syntactic Complexity*, F1; *Lexical Richness*, F4; *Repetitiveness*, F5; *Concreteness*, F6; *Positive Affect*, F8; and *Structural Variability*, F10). Of the remaining four, three shift from within 3 SE from the population mean to beyond 3 SE (*Lexical Sophistication*, F2; *Negative Affect*, F3; and *Conventionality*, F7), with the final factor remaining largely similar across both (*Narrative Drift*, F9).

For General Audiences fanfiction, we would thus characterize the style as leaning positive and avoiding negative topics, while having a quite variable writing style as it pertains to both individual word choice and overall structure. For Explicit fanfiction, we instead see less focus on varied word choices and structural diversity in favor of a high degree of concreteness and interrelated sentences leading to greater complexity. In other words, Explicit fanfiction seems focused on giving specific

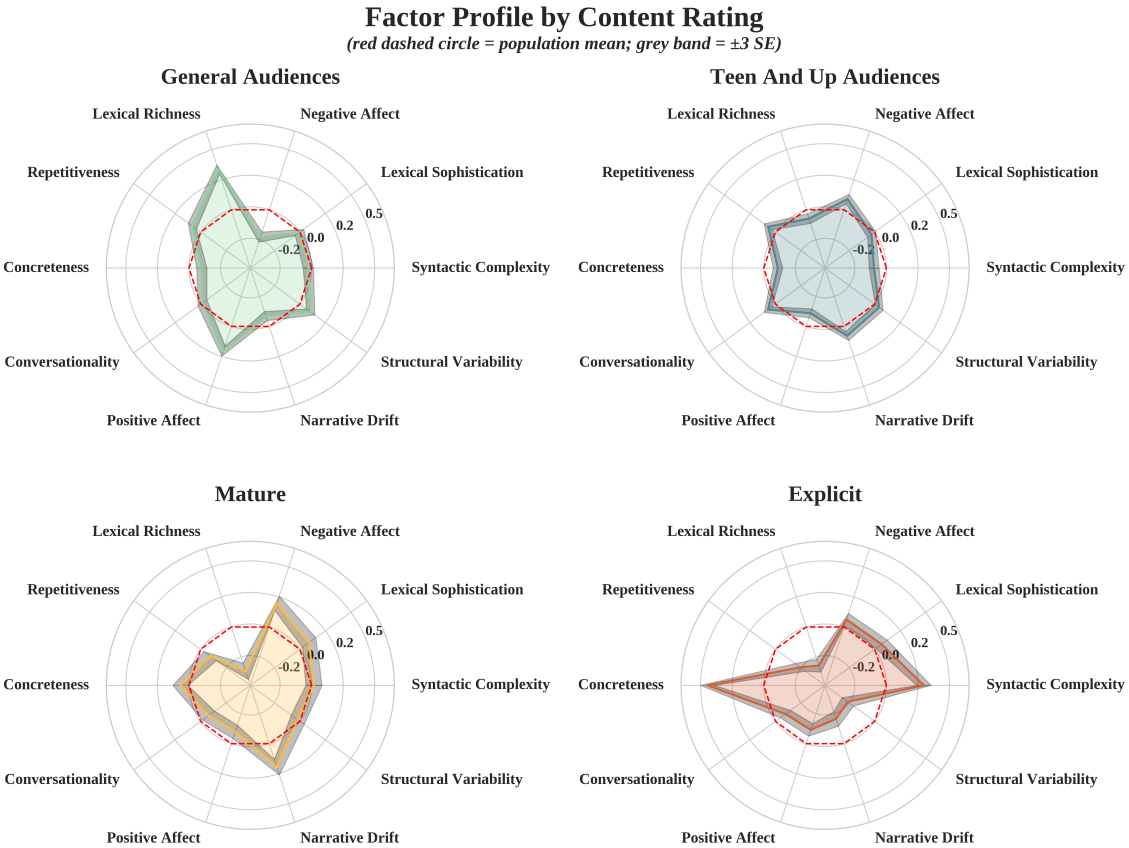


Figure 5: Mean factor-score profile by rating (10 EFA factors; grey band = ± 3 SE). Outside it is above the population mean; inside is below.

descriptions that are clearly organized, while General Audiences texts have a greater focus on rich and varied prose. This corresponds to previous findings for the linguistic style of different Maturity Rating, which finds Explicit fanfiction to have quite a distinct linguistic style [22].

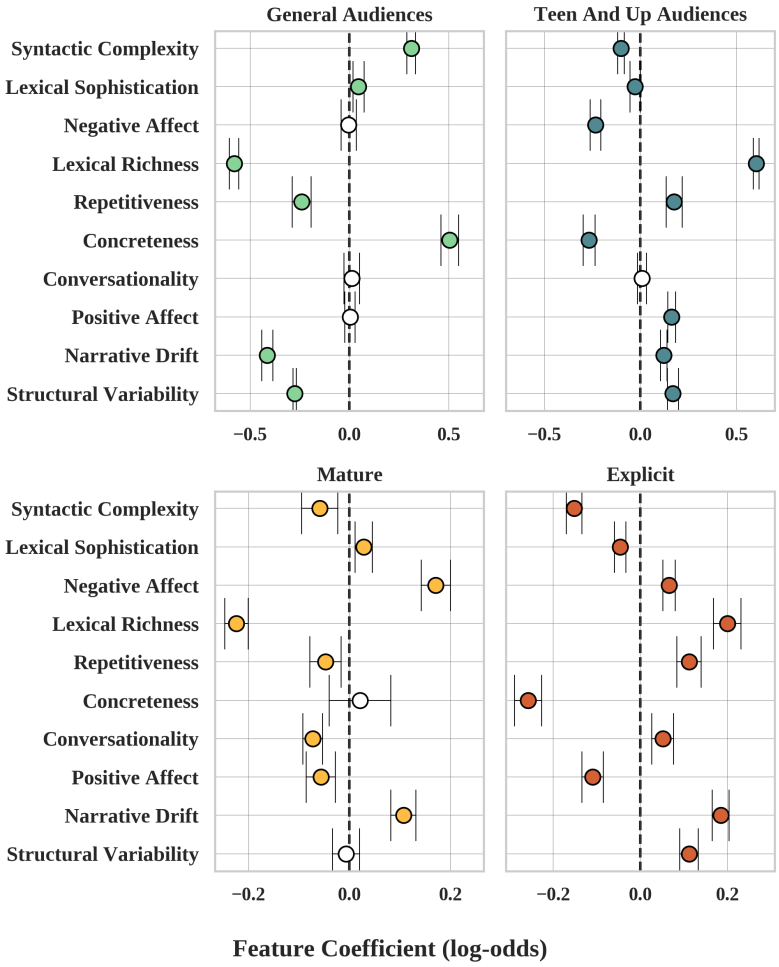


Figure 6: Coefficient plot from multinomial logistic regression predicting Maturity Rating from 10 EFA-derived stylistic factors. Dots show mean coefficients (log-odds) across 5 cross-validation folds. Filled dots indicate the mean is 3 or more standard deviations from the mean, while hollow dots indicate the effect is not stable across folds. Error bars are ±3 SD across folds.

However, in contrast to those findings, we find General Audiences to be more distinct than the three other ratings. This is especially evident in the model coefficients plotted on figure 6 (for a table with model coefficients see table 5 in appendix C). Here, General Audiences shows factor coefficients that are consistently the opposite sign of the three other ratings, most clearly seen in factors 1, 4, 5, 6, 9, and 10.

Meanwhile, Teen and Mature ratings quite closely follow the coefficients for Explicit, with notable exceptions only being *Conversationality* (F7) for Mature (β = -0.07), and both *Negative Affect* (F3) and *Positive Affect* (F8) for Teen and Up Audiences (F3 β = -0.23; F8 β = 0.16). Similarly for the factor-score profiles (on figure 5), both Mature and Teen and Up Audiences are quite close to the mean of the population, especially Teen and Up. This pattern could be an indication that the definitions of these in-between ratings are more subjective in their interpretation, and as such, they exhibit a less distinct linguistic style that makes them harder to distinguish, General Audiences and Explicit are quite clear-cut in their interpretation, constituting either end of a spectrum.

This may also relate to findings from the performance of the fandom model because, like the best-predicted fandoms, the two polar ratings have one or two clear, dominant factor(s) that uniquely signify writing for General Audiences or Explicit texts, respectively. The other two rating categories, though comparably unique in overall profile, are less interpretable and thus harder to classify.

It is worth noting that the model coefficients do present some surprising patterns. For example, with *Lexical Richness* (F4) for General Audiences texts, the model has a negative coefficient (β = -0.58), which is the opposite of what you would expect when looking at the factor-score profile for General Audiences. We see this pattern across factors and ratings, where it seems the model is not quite fitting accurately to the data, and so the model coefficients become skewed from what we would otherwise expect. Along with the poor model performance, it points to the fact that Maturity Ratings might not differ that much across these 10 factors, meaning the factors might not carry enough explainable power to drive an accurate classification model. This is surprising given the performance for the fandoms in RQ1 and previous findings showing a clearer linguistic distinction between Maturity Ratings [22]. More research is needed to assess what might be driving these results, given the distinct visual factor-profiles.

5 Limitations

The fandoms represented in this study – apart from RPF and Random, which could not be standardized as they were randomly sampled – are all based on works popular in western media. Additionally, due to constraints with the stylometric tools, we only study English-language texts. This begs the question as to whether the differences and temporal dynamics of authorial voice identified using these fandoms would present similarly in fandoms of works from differing cultural contexts and languages or not. In the case of this study, this is a question that we cannot address without broader analysis but is a relevant focus for future work.

The EFA of the chosen stylometric battery carries with it its own limitations. Because we used oblim rotation, the factors are correlated rather than independent in the EFA, mirroring their interrelatedness in real life. However, this means that coefficients attributed to one factor can partly reflect another rather than representing a fully distinct latent construct in the text. Additionally, the 10 factors together explain only 54% of total variance, thus not perfectly accounting for all of the systematic variation in the included stylometric scores. These limitations are important to highlight because factor scores produced in the EFA should be used to structure and guide interpretation rather than be treated as exact, directly measured quantities.

Additional considerations regarding the modeling of EFA factors, fandoms, and ratings also arose. For both the Fandom and Maturity Rating models, we debated the inclusion of repeated measures as we were concerned that there were individuals authoring works across fandoms that, if unaccounted for, could impact the model's predictions. This quickly grew complicated after considering the likelihood that some authors might write under several different pseudonyms, or use the possibility of publishing their texts anonymously without any authorship attribution. As it became clear that it was nearly impossible to control for repeated measures, we decided to forgo improved efficacy of the model for a more accurate representation of the entire corpus.

We also decided against the further optimization of the multinomial logistic regressions with the inclusion of possible random variables or further interaction effects. As stated before, this would require identifiers of individual text that we could trust to accurately represent those works, which we do not possess. Moreover, this would increase the unnecessary risk of the model overfitting the factors that already carry their own risk of multicollinearity, when we are primarily interested in the baseline coefficients, not predictive power.

6 Conclusion

In this paper, we have investigated how expressive profiles in fanfiction communities might differ across fandoms, as well as how they arise and to what degree they are influenced by genre conventions. By using 10 EFA factors derived from 71 stylistic measures, we have scored a corpus of almost 17,000 texts and have assessed how different groups can be classified based on these scores. We find that a multinomial logistic regression performs adequately at predicting a text's fandom from the 10 factors; however, Maturity Ratings provide more of a challenge. For the fandoms, we find individual fan communities to have distinct writing styles that are most easily separable when they have a set of few distinct factors. We also find that these writing styles emerge over time after a period of conventionalization, after which the fandoms fluctuate across a baseline style. There also seems to be no clear decrease in variance over time, though we see that the variance fluctuates along with the mean writing style. For Maturity Ratings, we similarly find that ratings with a few dominant features are more easily separable, which particularly pertains to General Audience texts and Explicit texts. We thus both replicate and introduce nuance to the findings from previous research with regards to the general writing style of fanfiction and the influential role of the community on how texts are produced over time.

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A Complete EFA Loadings

Metric	Category	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10
Question rate	Punctuation Style	−.32	–	–	–	–	–	+.54	–	–	–
Quotation rate	Punctuation Style	–	–	–	–	–	–	+.45	–	–	–
Exclamation rate	Punctuation Style	–	–	–	–	–	–	+.39	–	–	–
Ellipsis rate	Punctuation Style	–	–	–	–	–	–	+.33	–	–	–
Colon rate	Punctuation Style	–	–	–	–	–	–	–	–	–	–
Comma rate	Punctuation Style	–	–	–	–	–	–	–	–	–	–

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Table 6 – continued

Metric	Category	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10
Parenthesis rate	Punctuation Style	–	–	–	–	–	–	–	–	–	–
SD of sentence length	Document Scale	+.79	–	–	–	–	–	–	–	–	+.44
RIX	Readability	+.73	+.43	–	–	–	–	–	–	–	–
SMOG	Readability	+.50	+.62	–	–	–	–	–	–	–	–
Coleman-Liau	Readability	+.43	+.60	–	–	–	–	–	–	–	–
Duplicate 5-gram fraction	Repetition	–	–	–	−.61	+.54	–	–	–	–	–
Duplicate 8-gram fraction	Repetition	–	–	–	−.42	+.41	–	–	–	–	–
Top 2-gram fraction	Repetition	–	–	–	–	+.58	–	–	–	–	–
Top 3-gram fraction	Repetition	–	–	–	–	+.71	–	–	–	–	–
Top 4-gram fraction	Repetition	–	–	–	–	+.58	–	–	–	–	–
Duplicate line fraction	Repetition	–	–	–	–	+.43	–	–	–	–	–
Proportion of unique tokens	Lexical Richness	–	–	–	+.92	–	–	–	–	–	–
Alphabetic-token ratio	Lexical Richness	–	–	–	–	−.61	–	–	–	–	–
Out-of-vocabulary ratio	Lexical Richness	–	–	–	–	+.55	–	–	–	–	–
Syllables per token, mean	Word Form Complexity	–	+.90	–	–	–	–	–	–	–	–
Syllables per token, SD	Word Form Complexity	–	+.89	–	–	–	–	–	–	–	–
Token length, SD	Word Form Complexity	–	+.78	–	–	–	–	–	–	–	–
Age of acquisition, mean	Lexical Register	–	+.61	–	–	–	–	–	–	–	–
Windowed unigram perplexity, mean	LM Entropy	–	+.33	–	–	−.51	–	–	–	–	–
Windowed unigram entropy, SD	LM Entropy	–	–	–	–	+.42	–	–	–	–	–
Pronoun, prop.	POS Distribution	–	−.43	–	–	–	–	–	–	–	−.33
Coordinating conjunction, prop.	POS Distribution	–	−.37	–	–	–	–	–	–	–	+.31
Adverb, prop.	POS Distribution	–	–	–	–	−.37	−.34	–	–	–	–
Adjective, prop.	POS Distribution	–	–	–	–	−.36	–	–	–	–	+.31
Auxiliary, prop.	POS Distribution	–	–	–	–	–	−.78	–	–	–	–
Subordinating conjunction, prop.	POS Distribution	+.40	–	–	–	–	−.41	–	–	–	–
Particle, prop.	POS Distribution	–	–	–	–	–	−.40	–	–	–	–
Interjection, prop.	POS Distribution	–	–	–	–	–	–	+.59	–	–	–
Determiner, prop.	POS Distribution	–	–	–	–	–	–	−.50	–	–	–
Noun, prop.	POS Distribution	–	–	–	–	–	+.46	−.48	–	–	–
Adposition, prop.	POS Distribution	–	–	–	–	−.31	–	−.43	–	–	–
Verb, prop.	POS Distribution	–	–	–	–	–	–	–	–	–	−.58
Subordination ratio	Syntactic Complexity	+.97	–	–	–	–	–	–	–	–	–
Proportion of complex sentences	Syntactic Complexity	+.85	–	–	–	–	–	–	–	–	–
Parse-tree depth, mean	Syntactic Complexity	+.82	–	–	–	–	–	–	–	–	–
Coordination ratio	Syntactic Complexity	+.74	–	–	–	–	–	–	–	–	+.32
Parse-tree depth, SD	Syntactic Complexity	+.58	–	–	–	–	–	–	–	–	+.35
Dependents per head, mean	Syntactic Complexity	–	–	–	–	–	−.34	+.68	–	–	–
Proportion of adjacent dependency relations, SD	Dependency	−.54	–	–	–	–	–	–	–	–	–
Proportion of adjacent dependency relations, mean	Dependency	–	–	–	–	+.32	–	−.47	–	–	–
Dependency distance, SD	Dependency	–	–	–	–	–	–	+.42	–	–	+.62
Valence, SD	Affect Dimensions	–	–	+.75	–	–	–	–	+.48	–	–
Valence, mean	Affect Dimensions	–	–	−.58	–	–	–	–	+.56	–	–
Arousal, mean	Affect Dimensions	–	–	+.48	–	–	–	–	–	–	–
Dominance, SD	Affect Dimensions	–	–	+.35	–	–	–	–	–	–	–
Arousal, SD	Affect Dimensions	–	–	+.35	–	–	+.52	–	–	–	–
Dominance, mean	Affect Dimensions	–	–	–	–	–	−.40	–	+.37	–	–
Word concreteness, mean	Concreteness & Imagery	–	–	–	–	–	+.81	–	–	–	–

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Table 6 – continued

Metric	Category	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10
Word concreteness, SD	Concreteness & Imagery	–	–	–	–	–	+.75	–	–	–	–
Fear	Discrete Emotions	–	–	+.85	–	–	–	–	–	–	–
Anger	Discrete Emotions	–	–	+.83	–	–	–	–	–	–	–
Sadness	Discrete Emotions	–	–	+.79	–	–	–	–	–	–	–
Disgust	Discrete Emotions	–	–	+.77	–	–	–	–	–	–	–
Emotion diversity	Discrete Emotions	–	–	+.62	–	–	–	–	−.45	–	–
Joy	Discrete Emotions	–	–	–	–	–	–	–	+.85	–	–
Trust	Discrete Emotions	–	–	–	–	–	–	–	+.66	–	–
Anticipation	Discrete Emotions	–	–	–	–	–	–	–	+.57	–	–
Surprise	Discrete Emotions	–	–	–	–	–	–	–	–	–	–
Topic strength	Topic Strength & Volume	–	–	–	+.93	–	–	–	–	–	–
Semantic volume, log-determinant	Topic Strength & Volume	–	–	–	+.86	–	–	–	–	–	–
Maximum embedding-cluster share	Clustering	–	–	–	+.84	–	–	–	–	–	–
Semantic dispersion	Dispersion	–	–	–	–	–	–	–	–	+.66	–
Maximum pairwise semantic range	Coherence & Trajectory	–	–	–	–	–	–	–	–	+.79	–
Start-to-end semantic drift	Coherence & Trajectory	–	–	–	–	–	–	–	–	+.67	–
Segment transition variability	Coherence & Trajectory	–	–	–	–	–	–	–	–	+.59	–

Table 6: Factor loadings for all 71 stylometric features across the ten factors. Loadings within ±0.30 are ignored (shown as –). Each feature's primary factor is the one on which it loads strongest.

B Coefficients and Confusion Matrix for Fandom Model

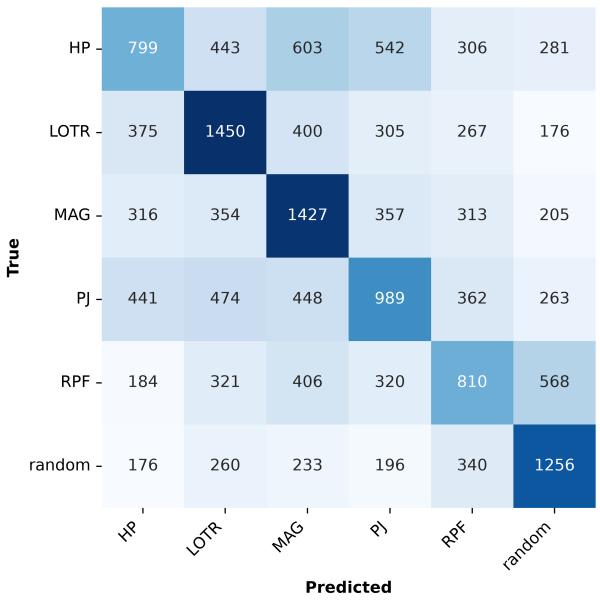


Figure 7: Confusion matrix for the 6-class Fandom classifier (5-fold CV, logistic regression on 10 EFA factors). Accuracy was 37.5%, with confusion spread fairly evenly off-diagonal.

F1: Syntactic Complexity	β	F2: Lexical Sophistication	β
Random	-0.47 ($\pm$ 0.022)	Random	0.31 ($\pm$ 0.016)
HP	0.24 ($\pm$ 0.011)	HP	0.22 ($\pm$ 0.014)
LOTR	0.17 ($\pm$ 0.009)	LOTR	-0.04 ($\pm$ 0.008)
MAG	0.05 ($\pm$ 0.007)	MAG	0.06 ($\pm$ 0.007)
PJ	-0.003 ($\pm$ 0.008)	PJ	-0.23 ($\pm$ 0.013)
RPF	0.02 ($\pm$ 0.015)	RPF	-0.32 ($\pm$ 0.010)
F3: Negative Affect	β	F4: Lexical Richness	β
Random	0.03 ($\pm$ 0.010)	Random	-0.19 ($\pm$ 0.013)
HP	0.05 ($\pm$ 0.009)	HP	-0.13 ($\pm$ 0.006)
LOTR	-0.09 ($\pm$ 0.009)	LOTR	-0.27 ($\pm$ 0.010)
MAG	-0.07 ($\pm$ 0.009)	MAG	0.05 ($\pm$ 0.008)
PJ	0.02 ($\pm$ 0.011)	PJ	0.08 ($\pm$ 0.008)
RPF	0.06 ($\pm$ 0.018)	RPF	0.46 ($\pm$ 0.008)
F5: Repetitiveness	β	F6: Concretness	β
Random	-1.17 ($\pm$ 0.017)	Random	0.34 ($\pm$ 0.011)
HP	0.43 ($\pm$ 0.006)	HP	-0.03 ($\pm$ 0.006)
LOTR	0.43 ($\pm$ 0.004)	LOTR	-0.13 ($\pm$ 0.010)
MAG	0.30 ($\pm$ 0.007)	MAG	-0.40 ($\pm$ 0.003)
PJ	0.44 ($\pm$ 0.008)	PJ	-0.01 ($\pm$ 0.008)
RPF	-0.42 ($\pm$ 0.008)	RPF	0.23 ($\pm$ 0.004)
F7: Conversationality	β	F8: Positive Affect	β
Random	0.40 ($\pm$ 0.014)	Random	0.12 ($\pm$ 0.009)
HP	-0.03 ($\pm$ 0.009)	HP	-0.03 ($\pm$ 0.008)
LOTR	-0.76 ($\pm$ 0.009)	LOTR	0.38 ($\pm$ 0.011)
MAG	-0.03 ($\pm$ 0.014)	MAG	-0.75 ($\pm$ 0.006)
PJ	0.10 ($\pm$ 0.007)	PJ	0.21 ($\pm$ 0.007)
RPF	0.32 ($\pm$ 0.012)	RPF	0.06 ($\pm$ 0.009)
F9: Narrative Drift	β	F10: Structural Variability	β
Random	-0.06 ($\pm$ 0.009)	Random	-0.30 ($\pm$ 0.012)
HP	0.11 ($\pm$ 0.008)	HP	-0.24 ($\pm$ 0.008)
LOTR	-0.09 ($\pm$ 0.012)	LOTR	0.38 ($\pm$ 0.007)
MAG	-0.10 ($\pm$ 0.007)	MAG	0.33 ($\pm$ 0.016)
PJ	0.34 ($\pm$ 0.011)	PJ	-0.14 ($\pm$ 0.006)
RPF	-0.20 ($\pm$ 0.003)	RPF	-0.03 ($\pm$ 0.009)

Table 7: Coefficient for each EFA factor across fandoms. Mean ($\pm$ standard deviation) across folds.

C Coefficients and Confusion Matrix for Ratings Model

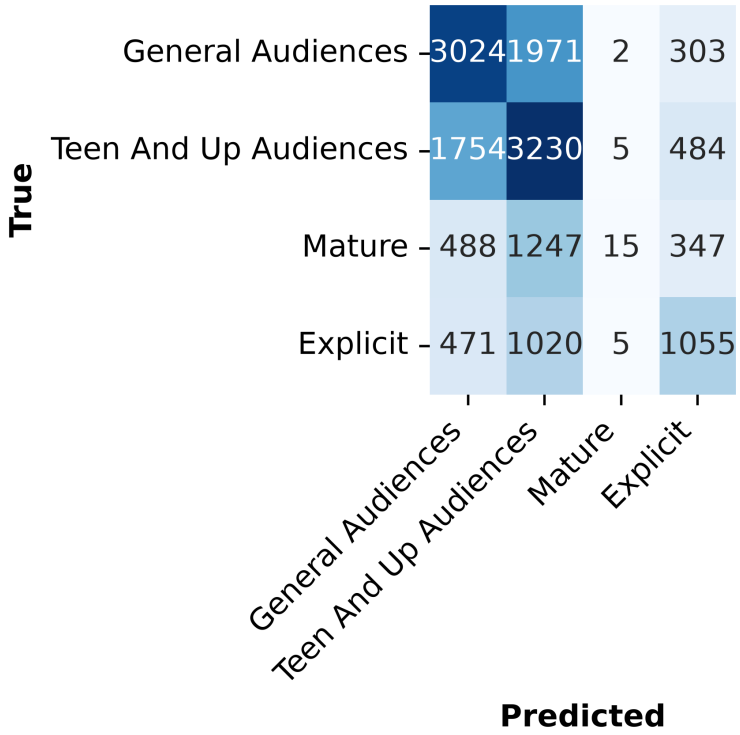


Figure 8: Confusion matrix for the 4-class Maturity Rating classifier (5-fold CV, logistic regression on 10 EFA factors). Accuracy was 47.5%, with Mature rarely predicted correctly.

F1: Syntactic Complexity	β	F2: Sentiment/Emotionality	β
General Audiences	0.31 ($\pm$ 0.007)	General Audiences	0.05 ($\pm$ 0.009)
Teen and Up Audiences	-0.10 ($\pm$ 0.006)	Teen and Up Audiences	-0.03 ($\pm$ 0.009)
Mature	-0.06 ($\pm$ 0.012)	Mature	0.03 ($\pm$ 0.006)
Explicit	-0.15 ($\pm$ 0.006)	Explicit	-0.05 ($\pm$ 0.004)
F3: Negative Affect	β	F4: Lexical Richness	β
General Audiences	-0.004 ($\pm$ 0.013)	General Audiences	-0.58 ($\pm$ 0.008)
Teen and Up Audiences	-0.23 ($\pm$ 0.009)	Teen and Up Audiences	0.60 ($\pm$ 0.005)
Mature	0.17 ($\pm$ 0.010)	Mature	-0.22 ($\pm$ 0.008)
Explicit	0.07 ($\pm$ 0.005)	Explicit	0.20 ($\pm$ 0.011)
F5: Repetitiveness	β	F6: Concretness	β
General Audiences	-0.24 ($\pm$ 0.016)	General Audiences	0.50 ($\pm$ 0.015)
Teen and Up Audiences	0.18 ($\pm$ 0.014)	Teen and Up Audiences	-0.27 ($\pm$ 0.010)
Mature	-0.05 ($\pm$ 0.010)	Mature	0.02 ($\pm$ 0.020)
Explicit	0.11 ($\pm$ 0.009)	Explicit	-0.26 ($\pm$ 0.010)
F7: Conversationality	β	F8: Positive Affect	β
General Audiences	0.01 ($\pm$ 0.013)	General Audiences	0.003 ($\pm$ 0.009)
Teen and Up Audiences	0.01 ($\pm$ 0.008)	Teen and Up Audiences	0.16 ($\pm$ 0.007)
Mature	-0.07 ($\pm$ 0.007)	Mature	-0.06 ($\pm$ 0.010)
Explicit	0.05 ($\pm$ 0.008)	Explicit	-0.11 ($\pm$ 0.008)
F9: Narrative Drift	β	F10: Structural Variability	β
General Audiences	-0.41 ($\pm$ 0.009)	General Audiences	-0.28 ($\pm$ 0.003)
Teen and Up Audiences	0.12 ($\pm$ 0.006)	Teen and Up Audiences	0.17 ($\pm$ 0.009)
Mature	0.11 ($\pm$ 0.008)	Mature	-0.01 ($\pm$ 0.009)
Explicit	0.18 ($\pm$ 0.006)	Explicit	0.11 ($\pm$ 0.007)

Table 8: Coefficient for each EFA factor across Maturity Ratings. Mean ($\pm$ standard deviation) across folds.